Week 7.1: Map

Introduction to Algorithms and Data Structures

SAXION UNIVERSITY OF APPLIED SCIENCES

We would like some (anonymous) feedback on this module!

edu.nl/acpfv

Map

- **Discuss work last lesson**
- Map introduction
- Map important concepts
- Summary
- Homework

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

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How to retrieve Teachers on first name?

(instead of on an index)

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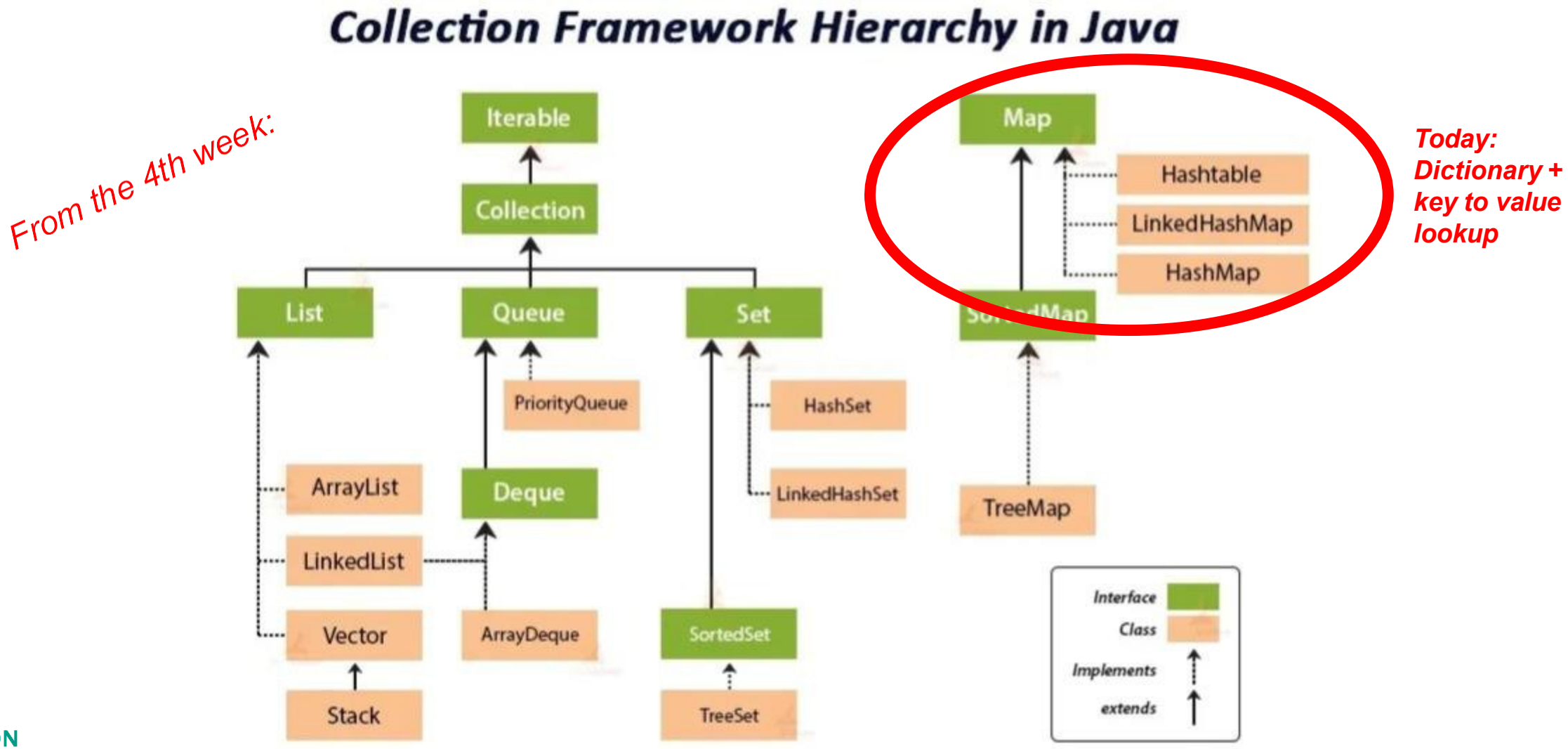
Remco?

Manasse?

Folkert?

JCF: the Java Collections Framework



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Array(List) versus Map (to store Teachers)

Map is an interface which uses Generic Types

You decide what type of keys you want to use.

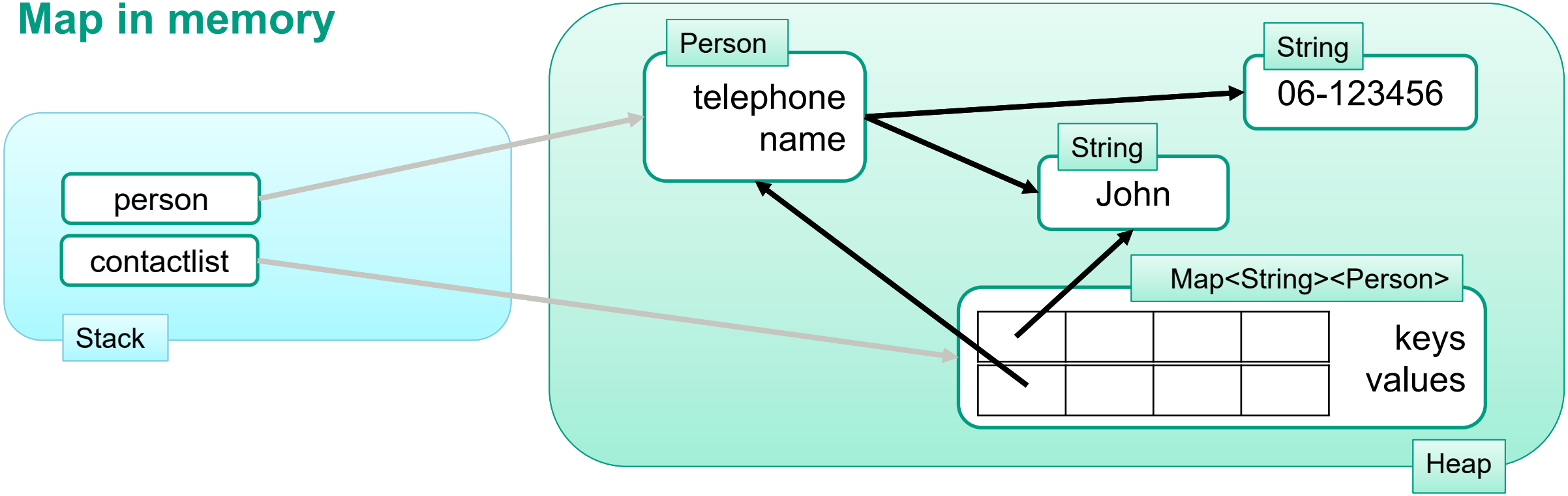
You decide wat type of values you want to store

HashMap is a Map implementation

Map is an interface

```
Map<String,Person> contactlist = new HashMap();
...
contactlist.put("john", new Person "john", "06-123456"));
...
Person j = contactlist.get("john");
```

Map in memory




```
Map<String,Person> contactList = new HashMap();
...
contactlist.put("John", new Person "John", "06-123456"));
...
Person person = contactlist.get("John");
```

What datastructure would you use?

Inventory Management System:
You are developing an inventory management system for a store where products need to be tracked, and each product has a unique (bar)code (number).

Contacts App:
You are tasked with developing a simple contacts application, which stores contacts.

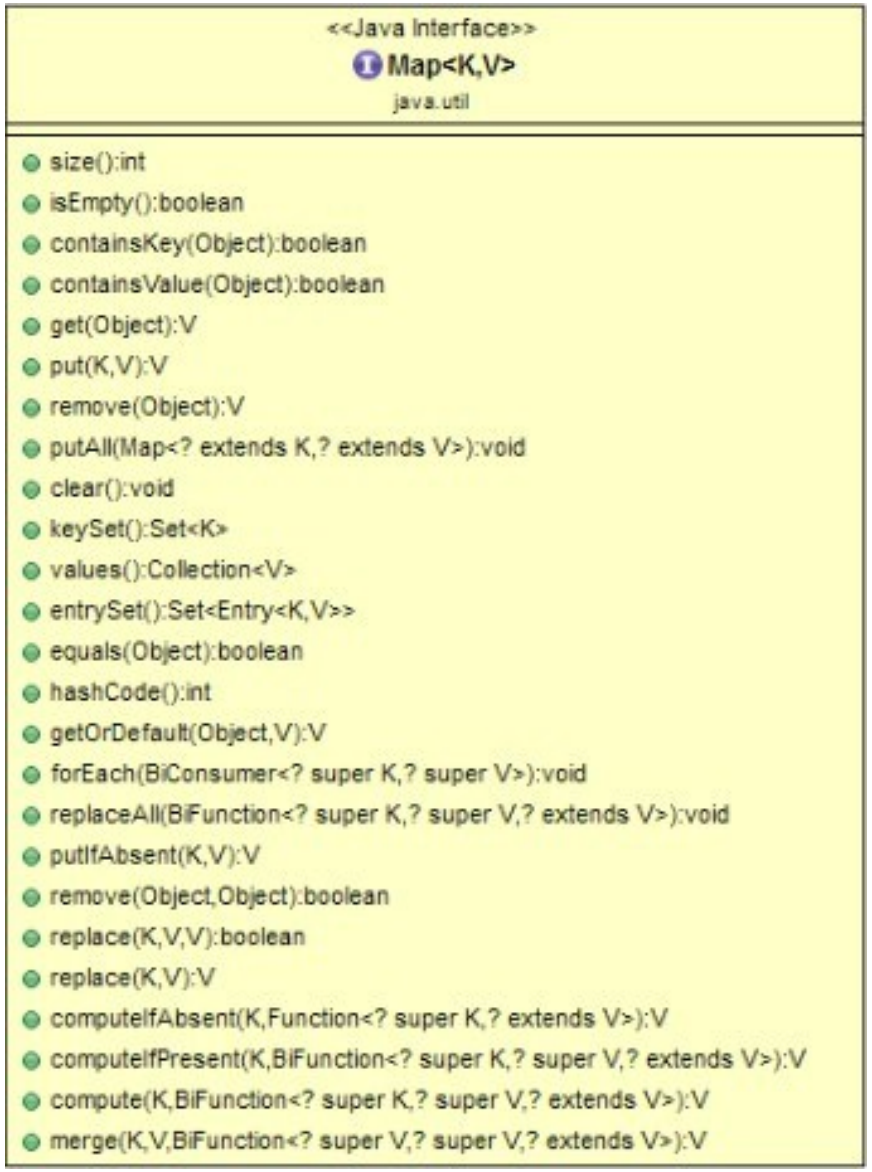
Student Grades Tracking:
You are creating a system to track and calculate grades for students, who have a unique student number.

Library Catalog System:
You are designing a catalog system for a library where books need to be managed.
Each book has a unique ISBN number.

Map(interface) methods:

- *"All the usual collection utility methods".*
- Notable mentions:
    - **put**(key, value) : *adding/replacing*
    - **get**(key) : *retrieving*
    - **keySet**() : *get all keys (= used search terms)*
    - **values**() : *get all values*
    - **replace**(key, value) : *strictly replace, (only) if k exists*
    - **replace**(key, old, new) : *(only) if get(k) == old: put(k,new)*

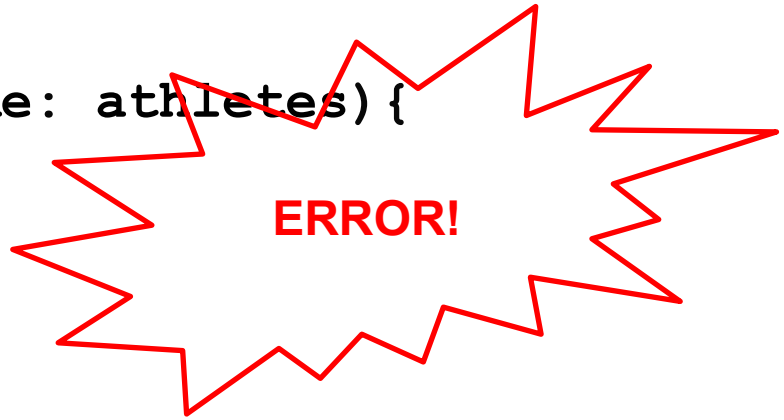
Which methods to use to mimmic (the database) **C**reate**R**ead**U**pdate**D**elete functionality?



Iterating over Map?

```
Map<Integer, String> athletes = new HashMap<>();
athletes.put(860, "Elaine Longjump");
athletes.put(1521, "Max Shortstep");
athletes.put(34, "Rico HeavyLift");

for(String name: athletes){
```



Iterating over Map: two options

Option 1: iterate over values

```
for(String name: athletes.values()){
  System.out.println(name);
}
```

```
Max Shortstep
Rico Hari
John Depp
```

Option 2: iterate over keys

```
for(int nr: athletes.keySet()){
  System.out.println("Nr: " + nr + " " + athletes.get(nr));
}
```

```
Nr: 1521 Max Shortstep
Nr: 34 Rico Hari
Nr: 860 John Depp
```

both cases:
no guarantees regarding order

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Summary of Map

- Map is an interface, which is
  - part of Java's Collection Framework (JCF) and
  - HashMap is the implementation you'll use most of the times.
- Map is a dictionary / associated array;
  - it uses a key (of any class type) to access
  - values (of any class type).

```
Map<String,Person> contactList = new HashMap();
...
contactlist.put("John", new Person "John", "06-123456"));
...
Person person = contactlist.get("John");
```

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Now it’s your turn!